NAME: _____ NetID:_____

INFO 4220 – Networks II: Market Design
Spring 2023 – Prelim 1

Instructions:

- **Put your name and NetID in every page (do it now). Exams will be scanned and processed using gradescope. If you don't put your name and NetID in the designated spaces, we may never be able to link your answers to you.**
- **Please do not separate the pages. If you do, staple them back at the end**
- **Time limit:** You have **60 minutes** to complete this exam.
- **Asking questions:** Read all the questions now and make any questions you may have at the beginning to avoid disturbing others. If you think something is wrong or some information is missing. Make the assumptions that you feel are necessary to answer, and state them along with your answer.
- **Cheat Sheet:** Only this printed exam and a 1-page cheat sheet are allowed in your table during the exam. Only 1 side of the cheat sheet can be written (handwritten, not printed), your name and netid must be written on top of the cheat sheet, your cheat sheet must be your own creation, and you have to turn in the cheat sheet with the exam.
- **Devices:** You cannot use any device (cell phone, smart watch, calculator, computer, etc.) during the exam.
- **Academic Integrity:** Any academic integrity violation will result in an academic integrity hearing in which a penalty consistent with Cornell's code of academic integrity will be determined.
- **Write clearly and show all your steps.** If you need more space for a question, use the last page.

Problem 1 (15 points, 3 points for each question)

1.1. In the context of the school choice problem, what is the difference between efficiency and stability? In which cases would you favor an algorithm that finds efficient assignments over an algorithm that chooses stable assignments?

Efficiency means that there is no other assignment u' in which all individuals are indifferent between u and u' and at least one person prefers u to u'.

(1.25 points)

Stability means that the assignment u is individually rational, there is no justified envy, and it is not wasteful.

(1.25 point)

The decision about which system to use comes down to the incentives and behaviors of schools. In a public school system we should care more about students getting the best possible assignment, and thus efficiency should be preferred as it doesn't leave any opportunity for students' welfare improvements unrealized. Stability should be preferred when there if schools have incentives to prefer

some students over others, and have ways of manipulating who gets admitted. If there is no evidence or opportunity for strategic behavior from schools, then efficiency should be preferred.

(0.5 points)

1.2. For the following game

	Player B		
Player A		Left	Right
	Left	(6, 6)	(0, 8)
	Right	(8, 0)	(2, 2)

a. Identify all possible Nash Equilibrium(s) and prove they are a Nash Equilibrium (1 point)

If b plays left, A plays right.

If b plays right, A plays A plays right.

If A plays left, B plays right.

If A plays right, B plays Right.

The Nash Equilibrium (unique) is (Right, Right). **(1 points for finding the equilibrium).**

To prove that it is a N.E. there are two valid explanations:

- Use the definition of a N.E. and show that:

$$\pi_A(Right, Right) \geq \pi_A(Left, Right) \Leftrightarrow 2 \geq 0$$

$$\pi_B(Right, Right) \geq \pi_B(Right, Left) \Leftrightarrow 2 \geq 0$$

- Alternatively say that the strategies used by each player are best response to each other:

$$BR_A(Right) = Right$$

$$BR_B(Right) = Right$$

(0.5 points for explicitly showing a N.E.).

Note: An alternative explanation is that playing right is a dominant strategy for each player. However, this is not a valid explanation if they don't mention that every equilibrium in dominant strategies is also a N.E.

b. Check if any of the Nash Equilibrium(s) you identified in a) is pareto dominated by other outcomes.

The outcome for both players is 2, we check if any combination provides a higher outcome.

The NE is pareto dominated by (Left, Left)

0.5 points for saying that (Right, Right) is pareto dominated by (Left, Left)

1 points for explaining why it is pareto dominated (because both players would obtain greater utility in (Right, Right) than in (Left, Left)

1.3. In the game from 1.2, now there is a probability of p that the judge of the game will change the rules so that the outcome of the game will be as follow:

	Player B		
Player A		Left	Right
	Left	(3, 3)	(8, 0)
	Right	(0, 8)	(6, 6)

Player B is a VIP player so he will be fully aware of the rule change. Player A is a regular player and only know there is a probability p the outcome of the game will as in part 1.3 (and otherwise as in 1.2).

a. For what values of p should player A play Right?

Player B will play right in 1.2 and left in 1.3. **(0.5 points)**.

Player A should play Right if the expected payoff of playing Right is ≥ expected payoff playing Left.

Option 1 (with prob. P the game is as in 1.2, and prob (1-p) the game is as in 1.3.

$2*(1-p) \geq 3p$

$p \leq 0.4$

(1 points)

Options 2 (in case anyone got confused for they typo, this will be considered correct as well)

$2*p \geq 3(1-p)$

$p \geq 0.6$

(1 points)

Note: Both options are considered equally correct (and are the same thing)

b. What would constitute a Bayesian Nash Equilibrium in this game?

An equilibrium in this Bayesian game is pair of strategies profiles that specify the actions to be played by each player given the signal they receive, such that when played they obtain a greater or equal expected payoff than with any other possible actions. Thus, the response should specify what each player should do based on the information they have:

Option 1 (if using definition that game in 1.3 happens with probability p)

Player A:

If $p \leq 0.4$, play right, otherwise left.

Player B

Play right if game doesn't change and left otherwise.

(1.5 point)

Option 2 (if using definition that game in 1.3 happens with probability 1-p)

Player A:

If $0.6 \leq p$, play right, otherwise left.

Player B

Play right if game doesn't change and left otherwise.

(1.5 point)

Note: Both options are considered correct.

1.5. Remember Alice, Bob and Amy from PS2? They discussed and updated their preferences, and their boss finally devised the following lottery: They would draw a random number from 1 to 4. If the result is 1, Alice will go to Dallas, Bob to NYC, and Amy to LA; if the result is 2, Alice will go to Dallas, Bob to LA, and Amy to NYC; and if the result is 3 or 4, Alice will go to NYC, Bob to LA, and Amy to Dallas. What is the probabilistic assignment that led to this lottery?

We use the following order:

	Dallas	NYC	LA
Alice			
Bob			
Amy			

We can infer the 3 deterministic assignments:

$$\frac{1}{4}\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}+\frac{1}{4}\begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 1\\ 0 & 1 & 0\end{pmatrix}+\frac{1}{2}\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{pmatrix}$$

(1.5 point)

NOTE: Be careful with the different ways/orders in which responses may write the order of people/cities. Consider the answer correct as long as the results are consistent with the stated order.

From there we can calculate the probabilistic assignment

$$\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{4} & \frac{3}{4} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \end{pmatrix}$$

(1.5 point)

1.5. There are 3 individuals i_1, i_2, i_3 and 3 houses h_1, h_2, h_3. i_1 is an existing tenant living in h_1, and i_2, i_3 are new tenants. Instead of having just an ordered preference list, these individuals obtain a continuous utility for being assigned to a house according to the table below (for example i_1 obtains a utility of 3 if assigned to h_1, and a utility of 1 if assigned to h_3). If the houses are going to be assigned using a random serial dictatorship with squatting rights, what should i_1 do: Keep h_1 or participate in the lottery?

	h_1	h_2	h_3
i_1	3	4	1
i_2	4	3	1
i_3	3	4	1

As this is a random serial assignment, there 6 ways in which the individuals could be arranged to pick houses. We can outline all possible orders, determine the house i_1 would get, and then computer their expected utility from participating in the lottery (and compare to the utility of keeping their house)

Order	I1 assignment	Utility
1-2-3	h_2	4
1-3-2	h_2	4
2-1-3	h_2	4
2-3-1	h_3	1
3-2-1	h_3	1
3-1-2	h_1	3

(1.5 points for realizing this is a random serial dictatorship and all possible orders should be considered)

Expected utility:

$$E[U(i_1)] = \frac{4}{2} + \frac{1}{3} + \frac{3}{6} = \frac{17}{6}$$

(0.5 point for the reasoning of computing the expected utility)

H1 keeps the house since the Expected utility is less than the current utility (17/6 < 3).

(1 points for concluding they should keep the house)

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Problem 2 (16 points, 4 each letter)

In a medical residence matching program, there are 5 doctors and 3 hospitals. Each hospital has a capacity of 2 doctors. Suppose there are preferences of doctors and hospitals as below:

$\boldsymbol{d_1}$	$\boldsymbol{d_2}$	$\boldsymbol{d_3}$	$\boldsymbol{d_4}$	$\boldsymbol{d_5}$
$\boldsymbol{h_1}$	h_1	h_1	h_2	h_2
$\boldsymbol{h_2}$	h_2	h_2	h_1	h_1
$\boldsymbol{h_3}$	h_3	h_3	h_3	h_3

$\boldsymbol{h_1}$	$\boldsymbol{h_2}$	$\boldsymbol{h_3}$
d_5	d_3	d_3
d_3	d_4	d_4
d_4	d_2	d_5
d_1	d_1	d_1
d_2	d_5	d_2

a. If using the deferred acceptance algorithm in this problem, would you use doctors or hospitals proposing. Explain your choice.

Doctors propose because it is strategyproof **(4 points)** and Roth and Peranson determined that it provided fairer outcomes for candidates.

b. Use the algorithm you chose in (a) to find a solution to this matching problem.

Step 1:

D1 -> h1

D2 -> h1

D3 -> h1

D4 -> h2

D5 -> h2

D2 rejected.

(1 point)

Step 2: D2 -> h2, d5 rejected.

Step 3: D5 -> H1, d1 rejected.

(1 point)

Step 4: D1 -> h2, D1 rejected.

Step 5: D1 -> h3.

(1 point)

The assignment will be:

H1: {d3, d5}

H2: {d2, d4}

H3: {d1}

(1 point)

c. Is there any stable matching in which hospital 3 fills its vacancies? Explain your answer.

No, because according to the rural hospital theorem, if one hospital does not fill its vacancies in one stable matching, it will not fill them in any stable matching (alternatively, that if a hospital does not fill its vacancies in a stable matching, it is matched to the same set of doctors in all stable matchings).

Provide 4 points with the explanations.

d. After seeing their match, d_1 decides that they would rather be unemployed than accept the match they were assigned to and refuses to work for that hospital (let's say they quit on the first day). Will the resulting match (after doctor 1 quits) be stable? What if it is d_5 (and not d_1) that quits on day 1?

If d_1 quits, the matching will still be stable **(1 point)** d_3 will be empty, but no doctor would like to move to h_3 (which wasn't full to begin with), so the assignment will not be wasteful **(1 point)**.

If d_5 quits, the matching will no longer be stable **(1 point)** because it will be wasteful since h_1 will not be full and d_2 likes h_1 better than h_2 **(1 point)**.

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Problem 3 (19 points)

You have been eating pizza with three friends, but the number of slices was impossible to divide equally. Each of you could eat your preferred pizzas until there are only 2 slices left: All meat and vegetarian. You don't have a knife, so you cannot divide the pizza slices any further. To show off your knowledge about assignment algorithms, you tell your friends that the best way to decide who gets the last 2 slices of pizza is it to find a random assignment using the eating algorithm and then use a lottery that follows the random assignment to allocate the pizza slices. One of your friends is a vegetarian and doesn't eat meat. The other two friends are severely allergic to peppers (which the vegetables pizza contains), so they can only the meat pizza. You enjoy both options but slightly prefer vegetarian.

a. Find the assignment that results from using the eating algorithm if everyone eats at the same speed. (5 points).

You	Friend 1	Friend 2	Friend 3
Vegetarian	Vegetarian	All meat	All meat
All Meats	Friend 1	Friend 2	Friend 3

(1 point)

Step 1:

At time 0 You and Friend 1 start eating Vegetarian, and Friend 2 and Friend 3 start eating all meat. After some time, both pizzas are fully eaten and:

You ate ½ vegetarian.

Friend 1 ate ½ Vegetarian.

Friend 2 ate ½ All meat.

Friend 3 ate ½ All meat.

(2 points)

The probabilistic assignment is:

Person	Vegetarian	Meat	None
You	$\frac{1}{2}$	0	$\frac{1}{2}$
Friend 1	$\frac{1}{2}$	0	$\frac{1}{2}$
Friend 2	0	$\frac{1}{2}$	$\frac{1}{2}$
Friend 3	0	$\frac{1}{2}$	$\frac{1}{2}$

(2 points)

Note: If they did the algorithm and the result correct, they can get full points without explaining the preferences. Don't discount any points if column none is not provided.

b. If you are really kind of indifferent between vegetarian or meat (you prefer vegetarian, but for a very tiny margin), and thus what you truly want is to maximize your probability of getting "a" slice of pizza. Is the mechanism used in part (a) strategy-proof. Explain why or why not (5 points).

The key aspect of this question was to realize that your objective was to maximize the probability of getting any slice of pizza, not just your preferred slice of pizza. Also, note that Friends 1, 2, and 3 can only eat one type of pizza, so they will never lie about their preferences. Thus, all that needed to be considered in this question was to see if you had a larger probability of getting any slice of pizza by being truthful or by inverting your preferences.

The mechanism is not strategyproof, because there is a way in which you can increase your chances of eating by changing your preferences:

You	Friend 1	Friend 2	Friend 3
All Meats	Vegetarian	All meat	All meat
Vegetarian	Friend 1	Friend 2	Friend 3

(1 point)

We run the algorithm again:

Step 1:

At time 0 you, friend 2, and friend 3 start eating all meat. Friend 1 starts eating vegetarian.

After some time, the all meat pizza is full eaten. At this point:

You are 1/3 All Meat

Friend 1 ate 1/3 Vegetarian (and there is 2/3 of the vegetarian pizza left)

Friends 2 and 3 are 1/3 All Meat each

(1 point)

Step 2:

You start eating vegetarian, and friend 1 continues to eat vegetarian.

After some time, the vegetarian pizza is fully eaten and:

You are 1/3 Vegetarian

Friend 1 are 1/3 Vegetarian (plus 1/3 he had already eaten in previous step)

Friend 2 and Friend 3 spend 1/3 without eating.

(1 point)

The new assignment will be:

Person	Vegetarian	Meat	None
You	1/3	1/3	1/3
Friend 1	2/3	0	1/3
Friend 2	0	1/3	2/3
Friend 3	0	1/3	2/3

With truthful preferences: You have a 1/2 chance of getting "any" slice of pizza.

If misrepresenting preferences: You have a 2/3 chance of getting "any" slice. Thus, if objective is to eat "any" slice of pizza is better to lie about your preferences.

(2 points)

c. Does the assignment you found in part (a) satisfies the equal treatment of equals? Explain why or why not (4 points).

Yes, because friends 3 and 4 have the same preferences and get the same probabilistic assignment **(4 points).**

d. Your friends have decided that as you can eat both types of pizza, and enjoy both almost equally, you have an unfair advantage. Thus, they want you to eat at half their speed in the eating algorithm. Assuming, that all you want is "a slice of pizza" (not caring of which type), what preference ordering should you submit (5 points).

We compare both options with the new speed. To avoid double jeopardy, the score for doing the eating algorithm with different speeds correctly once (with the true preferences) carries more weight than repeating it with non-truthful preferences.

Option 1: Original (real) preferences.

Step 1

You and fried 1 start eating vegetarian, friends 2 and 3 start eating all meat. By the time the all meat pizza is over:

You: Ate 1/4 Vegetarian

Friend 1: Are 1/2 vegetarian.

Friend 2 and 3: Ate 1/2 All Meat.

(1 points)

Step 2

You and Friend 1 continue to eat vegetarian, while friends 2 and 3 have nothing to eat. By the time the vegetarian pizza is over

You: Are 1/12 Vegetarian (plus the 1/4 from the previous step).

Friend 1: Are 1/6 Vegetarian.

Friend 2 & 3: Don't eat anything extra.

(1 points)

The assignment for this case is: **(2 point for table, not necessary to include "none" column)**

Person	Vegetarian	Meat	None
You	1/3	0	2/3
Friend 1	2/3	0	1/3
Friend 2	0	1/2	1/2
Friend 3	0	1/2	1/2

Option 2: Fake preferences

Step 1

You, Friend 3, and 4 start eating all meat. Friend 1 start eating vegetarian. After 2/5 time, the all meat pizza is fully eaten. At that point:

You: Ate 1/5 All Meat.

Friend 1: Ate 2/5 Vegetarian.

Friend 2 & 3: Ate 2/5 All Meat.

(0.125 points)

Step 2:

You move to eating vegetarian, Friend 1 keeps eating vegetarian, friends 2 and 3 have nothing to eat. By the time the vegetarian pizza is fully eaten, each ate an additional:

You: 1/5 Vegetarian (plus 1/5 All meat from previous step)

Friend 1: 2/5 Vegetarian (plus 2/5 vegetarian from previous step)

Friend 2 & 3 don't eat anything extra.

(0.125 points)

The assignment for this case is:

Person	Vegetarian	Meat	None
You	1/5	1/5	3/5
Friend 1	4/5	0	1/5
Friend 2	0	2/5	3/5
Friend 3	0	2/5	3/5

(0.25 point)

You should still lie and submit that you prefer All meat over Vegetarian as $\frac{2}{5} > \frac{1}{3}$ **(0.5 point).**